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**Understanding System Limits and Lab/System-
Testing of HF-Receive-Antenna-Phasing-
Controllers per example DXE NCC-2/NCC-1**

Background, Device, Lab + System Tests

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Understanding System Limits and Lab/System-Testing of HF Receive Antenna Phasing Controllers per example DXE NCC-2/NCC-1

Motivation:

Electromagnetic Interference/disturbance has consistently grown for decades based on the proliferation of partly badly denoised electronic equipment. Today on HAM RADIO HF this is a seriously limiting factor in many residential areas. For details about EMI distribution investigation see Ph.D. Thesis 2023 of PA0KDF

<https://research.utwente.nl/en/publications/cumulation-of-man-made-radio-noise-a-quest-for-causes-of-the-incr/>

For EMI-Mitigation in the HAM shack see

<https://www.qrz.com/db/HB9CVQ> => breaking news, HAM RADIO 2026 Slides Download, EN-Version

Sorry, error correction at end of slide # 32: +6dBm => 113dBuV = 447mV 50Ohm

If all denoising measures are correctly implemented and there is still EMI/local QRM coming into the antenna from a problematic environment Noise Cancellation in the shack is one further option.

1. Historical Description

I would characterize the NCC-2 this way:

The NCC-2 is not the descendant of one single noise-canceller invention. It is the convergence of early phased-array null steering, auxiliary-reference interference subtraction, military adaptive cancellers, and amateur-radio QRM-eliminator circuits.

The basic physics has been used since the earliest decades of radio. Recognizable amateur auxiliary-antenna cancellers were documented by the 1970s, commercial QRM eliminators became established in the late 1980s and 1990s, and the NCC series developed the technique into a *higher-dynamic-range, wide-frequency, fully adjustable receiving-array controller*.

For typical urban evening and nighttime EMI HAM Radio band peaks, the crucial question is not simply whether *the auxiliary antenna* receives more noise. It is whether this antenna/sensor receives a *clean, stable reference copy of the same noise appearing on the main RX antenna*. (Important: Always suppress any common-mode conducted paths with ferrites/chokes!)

If it does, the NCC-2 should be capable of a substantial null/notch (Test Report chapter). If several unrelated sources contribute to the nighttime noise floor, the achievable reduction will be much smaller, and the null may move with frequency or time.

1.1 RF phasing (external) vs. modern DSP (inside the TRX)

External, before receiver frontend, interference reduction prevents overload.

“DSP noise reduction” covers several different technologies. They should not be treated as one function.

- Noise blanker

A noise blanker identifies short, high-amplitude impulses and suppresses or replaces the affected samples.

Best targets include:

- ignition impulses
- electric-fence pulses
- some power-line arcing
- repetitive switching spikes

It can be highly effective when the interference consists of isolated impulses. It is usually poor against continuous broadband hash or a dense noise comb. Strong signals inside a wide receiver passband can also trigger false blanking and create distortion.

- Adaptive noise reduction

Conventional noise reduction estimates which parts of the signal are wanted speech or CW, and which parts are uncorrelated background noise. Modern implementations may use time-domain adaptive filters, spectral subtraction or other statistical methods.

For example, FlexRadio documents a time-domain “Leaky LMS” noise reducer for random background noise, frequency-domain spectral subtraction for steady noise and speech-specific processing using voice-activity detection.

These functions can make received audio sound quieter and improve listening comfort. They do not necessarily increase the RF signal-to-noise ratio.

At low SNR, they may produce:

- watery or metallic audio
- musical-noise artifacts
- clipped consonants
- loss of weak CW elements
- apparent quieting without improved information recovery

- Automatic notch filter

An adaptive notch filter identifies a stable carrier, heterodyne or hum component and places one or more narrow notches on it. FlexRadio, for example, documents both adaptive time-domain and FFT-based notch functions for steady tones and narrow spectral peaks.

This is excellent for:

- an unmodulated carrier
- a tuner birdie
- a fixed switching spur
- mains-related hum
- a stable heterodyne

It does not remove broadband modulation surrounding the carrier, and it cannot spatially separate two stations occupying the same channel.

- Neural-network noise suppression

Newer systems can classify speech-like content and suppress complex noise using trained neural networks. E.g., FlexRadio's current SmartSDR documentation includes recurrent neural-network suppression intended for mixed or changing noise such as fans, equipment and electrical noise. RM-Noise <https://ournetplace.com/rm-noise/> via internet (latency) tries noise reduction similar techniques.

This can produce impressive speech intelligibility, but it remains content-based processing. It may be less appropriate for:

- weak CW
- digital modes
- non-voice signals
- measurements
- natural-sounding AM reception
- signals far below the interference when the ADC is already overloaded

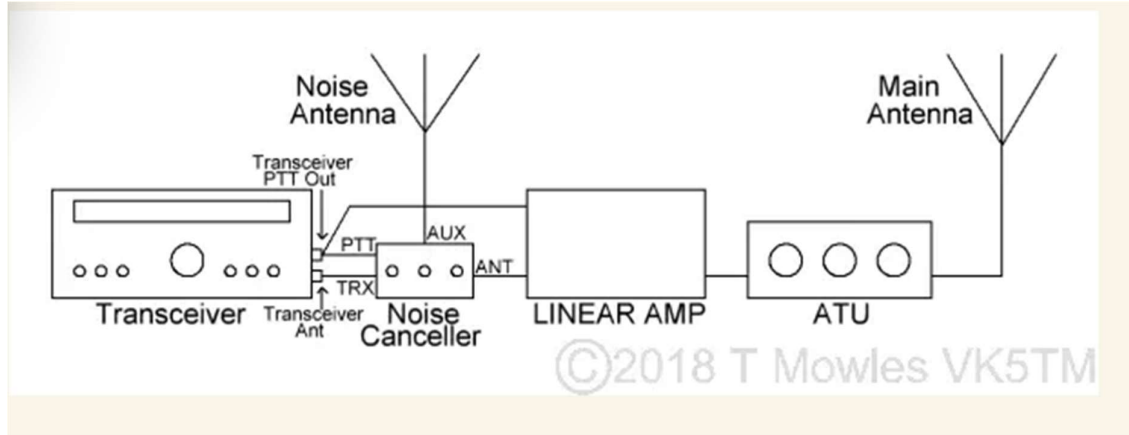
1.2 Closest known patent families

Patent / publication	Assignee / inventor	Why it is relevant
USH740H, "Antenna sidelobe interference canceller", published 6 Feb 1990	U.S. Army / Frank S. Gutleber	Very close conceptually: a primary receiving antenna and auxiliary antenna pair isolate interference, then adjust magnitude and phase so the interference is <i>equal and 180° out of phase</i> with the component in the primary antenna signal before subtraction. It is more military/array/jammer oriented than a simple HF QRM eliminator, but the cancellation principle is directly aligned.
USH739H, "Auxiliary antenna interference canceller", Priority 6 Feb 1990	U.S. Army / Frank S. Gutleber	Uses a high-gain primary receiving antenna plus a scannable auxiliary antenna array. Interfering signals are isolated, then separately phase and amplitude adjusted and fed to subtractors at the primary receiver output. Again, not specifically a small HF shack box, but highly relevant to auxiliary-antenna cancellation
US10469110B1 / US20190319651A1, "Noise canceller device", Priority 14 Sep 2016	Hitachi Kokusai Electric / Kokusai Denki	More modern and very relevant: a first/main antenna receives the desired signal, a second/sub antenna receives interference, and adjustment units <i>vary delay, phase, and amplitude</i> of the sub-antenna signal. The device searches for settings that minimize average power, then subtracts the optimized interference replica from the main antenna signal.
WO2018051758A1, "Noise canceller device" Priority 24 Aug. 2017	Hitachi Kokusai Electric / Kokusai Denki	International family member of the above noise canceller concept, explicitly classified under H04B1/126, which covers adaptive balancing using multiple inputs, for example an auxiliary antenna receiving the interfering sign.

US8428540B2, “Communication apparatus, noise canceller, noise cancelling method...” Priority 16 Aug. 2008	Fujitsu	Related RF receiver noise cancellation, using stored I/Q parameters and anti-phase radio-wave noise in addition to cancel internal radio noise. It is less like a manual HF auxiliary antenna phaser, but it is still in the same broad patent class of receiver noise cancellation using anti-phase RF correction.
US8422974B2 / US20100048156A1, “Noise cancellation for antenna module” Priority 20 Aug. 2009	Intersil Americas	Describes an antenna module with detector and noise canceller integrated near the antenna path, intended for EMI/spur/interference cancellation in devices such as GPS or wireless receivers. It is more integrated-electronics oriented than HF auxiliary antenna phasing.
US10476154B2, “Auxiliary antenna array for wideband sidelobe cancellation” Priority 05 Aug. 2015	MIT	Uses auxiliary antenna elements with variable attenuators, phase shifters, or true-time-delay units to cancel sidelobes in simultaneous transmit/receive systems. It is microwave/STAR systems rather than HF reception, but it is a strong modern example of auxiliary antenna cancellation with phase, amplitude, and delay control.

2. Basic Principle

A general set-up is shown by VK5TM (see also info X-Phaser, QRM eliminator, Phasing Network-Bridge PA0SIM 2019, like MFJ-1025 1997, ANC-4 2003, G4WIF 2020) in the following sketch.



The NCC-2 belongs to a long line of *two-channel interference-cancellation* and antenna-array (minimum 2 antennas, one main and one small auxiliary, interference pick-up) techniques. The underlying principle is old, but the NCC-2 (after 2016) is a particularly refined, improved *analogue* implementation designed for demanding MF/HF reception. The previous model was NCC-1.

Assume that the main antenna (CH A) receives a wanted DX signal S plus local interference N :

$$V_A = S_A + N_A$$

A second antenna (CH B) receives another version of the same interference:

$$V_B = S_B + N_B$$

The second signal is adjusted in amplitude and phase by a complex weighting factor w , then added to the first:

$$V_{\text{out}} = V_A + wV_B$$

The controls are adjusted so that:

$$N_A + wN_B \approx 0$$

This requires two simultaneous conditions:

1. The *amplitudes* of the two noise samples *must be equal*.
2. *Their phases must differ by approximately 180°* at the combiner in the NCC-2.

The wanted signal is preserved only to the extent that it has a different amplitude and phase relationship between the two antennas:

$$S_A + wS_B \neq 0$$

Therefore, the system does not identify “noise” in the way a DSP algorithm might. It simply cancels whatever signal component has the selected spatial amplitude and phase relationship. That component can be unwanted RFI, EMI, power-line noise, an inverter, a switching supply, or even an unwanted radio station.

The key success factor is the "noise" signals (CH A/CH B) need to be correlated!

CH A sees a noise signal copy of CH B.

Phase differences can only be adjusted within instrument limits (Test Report). If out of range a coax delay line may help to shift the phase into typical 0-180 deg. Switching Norm/Rev or H/L may also sometimes more easily help.

The biggest difference to other phase controllers on the market is the wide total independence of amplitude and phase-control adjustment.

We must distinguish however between Near-Field (NF) -very roughly Wavelength/6, mainly inductive/capacitive coupling, almost no emanating radiation and Far-Field (FF).

Here for a full halfwave dipole the relevant distance starts typically around 2 to 4 wavelengths. Size of the Radiator vs. Wavelength matters. The smaller the size the earlier FF can be reached. Here the free space impedance is $Z_0 = E/H = 377 \text{ Ohm}$ (Plane Wave, TEM). E-field (Voltage/distance, V/m) and H-field (current/distance, A/m) follow so to speak Ohms-Law for vacuum. Antenna gain also matters. A high gain antenna requires more distance to meet FF and form the true radiation pattern.

2.1 Near/Far-Field

At HF, a simple rule of thumb for the reactive near field is roughly:

$$r \lesssim \frac{\lambda}{2\pi}$$

Within this region, stored electric and magnetic fields can dominate and nearby objects can alter antenna impedance, pickup, and coupling behavior. Near-field regions are

commonly separated into reactive near field, radiating near field, and far field, with the far field often beginning around $2D^2/\lambda$ for an antenna of largest dimension D .

For HF: (for 160m, 1.8MHz, $\lambda/6 \Rightarrow$ ca. 28m)

Band	Frequency	Wavelength	Approx. $\lambda/2\pi$
80 m	3.5 MHz	86 m	13.7 m
40 m	7 MHz	43 m	6.8 m
20 m	14 MHz	21 m	3.4 m
10 m	28 MHz	10.7 m	1.7 m

So, **if** your EMI source is only about **5 m** from the noise antenna, it can be very much in the near-field zone in the 80 m and 40 m band, and still quite relevant on 20 m.

Hint: The “small movement” Test

This is the simplest and most revealing test.

Procedure

1. Tune to the offending noise.
2. Set the NCC-2 for the deepest null.
3. Mark the noise antenna position.
4. Move the noise antenna in small steps:
 - 20 cm
 - 50 cm
 - 1 m
5. At each position, record:
 - NCC-2 phase setting
 - NCC-2 balance setting
 - null depth
 - received noise level on noise antenna alone
 - wanted signal level on noise antenna alone

- whether the null is stable

Interpretation

Likely near-field coupling if:

- 20 to 50 cm movement changes the null dramatically
- the phase setting jumps a lot
- the balance setting moves from one extreme to another
- the noise level changes by many dB over a very small movement
- touching/moving nearby objects changes the null

More likely far-field/radiated pickup if:

- changes are smooth with distance
- null remains repeatable
- phase setting shifts gradually
- rotating (small loop) the antenna gives logical pattern changes

Near-field coupling is often dominated by local electric or magnetic fields and can be strongly affected by nearby conductive objects, wiring, masts, and ground paths.

2.2 The feedline sensitivity test (Common Mode Test)

This one is very important. Often the “noise antenna” is not the real antenna. The coax shield is.

Procedure

With the noise antenna fixed:

1. Set a good null on the NCC-2.
2. Move only the coax feedline by hand or with an insulated rod.
3. Lift the coax away from the ground.
4. Move it closer/farther from house wiring.
5. Coil/uncoil 1 to 2 m of coax.
6. Add a common-mode choke near the antenna feed point.
7. Add another choke near the shack entry.

Interpretation

Strong near-field/common-mode coupling if:

- moving the coax cable changes the noise level
- moving the coax changes the NCC-2 phase setting
- touching the coax shield or connector changes the null
- adding a choke changes the null depth or phase setting significantly
- The loop rotation has little effect, but coax movement has a big effect.

That usually means the feedline is part of the pickup system.

Good sign: after adding CM-chokes, the antenna rotation (loop) and placement become more logical and repeatable.

2.3 The “wanted signal loss” test

This tells whether your noise antenna is too good as a DX antenna.

Procedure

1. Pick a stable wanted signal.
2. Listen on main antenna alone.
3. Listen on noise antenna alone.
4. Set NCC-2 for maximum noise cancellation.
5. Note how much the wanted signal drops.

Interpretation

Good noise antenna placement:

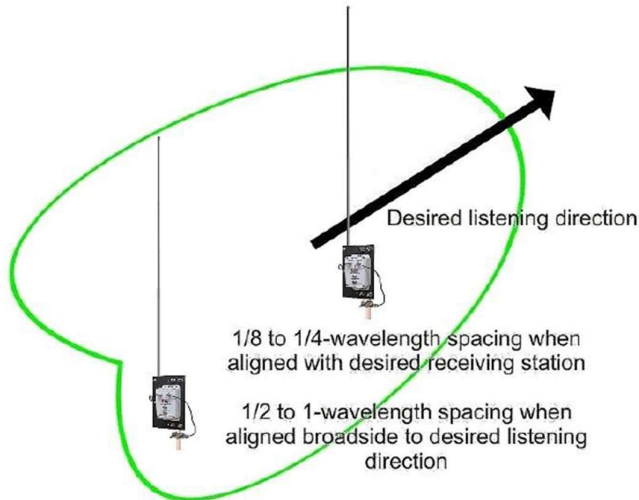
- Noise drops a lot
- wanted signal drops little
- Readability improves

Poor placement:

- noise drops
- wanted signal also drops strongly
- audio may sound hollow or phased
- some stations disappear at the noise null setting

That means the noise antenna is hearing too much of the wanted signal in a cancellable relationship.

2.4 Principle of RX-Array Phasing



(Source DXE NCC-2 Manual)

This set-up may be for low band weak RX signal operation too insensitive based on limited antenna rod length despite pre-amps. Remember own Tx coupling must be well controlled (limiter mostly work fine for low input impedance). Combining Antennas to Improve Signal-to-Noise Ratio (ant. pattern) is one option to control some EMI/RFI. Small magnetic loops are excellent in local noise minimizing (90 deg. rotation).

SDR diversity reception: *two coherent receivers* can do adaptive nulling, similar in concept to phasing but sometimes easier to visualize and optimize.

The following table shows for **RG58C/U coax cable** the **phase shift** for selected cable length and classical HAM Bands. The **time delay** equivalence (ns, group delay) for a 1m RG58C/U (velocity factor 0.66) is ca. 5ns/m. This could be used for extended beam shifting of RX antenna arrays or phase alignment range in front of NCC-2.

Even a 0.5m short piece of coax cable can shift phase on 20m more than 10deg.

Phase shifts for RG58C/U ($v-f=0.66$) coax cable

Table of phase shifts (degrees)

- **Format in each cell:** *Total degrees; Equivalent phase (mod 360°)*

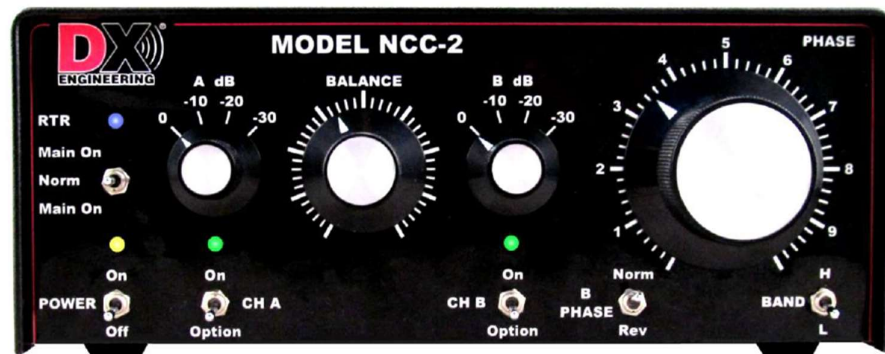
Length	1.820 MHz	3.520 MHz	7.020 MHz	14.020 MHz	21.020 MHz	28.020 MHz	50.120 MHz
10.0 m	33.1°; 33.1°	64.0°; 64.0°	127.6°; 127.6°	255.6°; 255.6°	383.6°; 23.6°	511.6°; 151.6°	911.3°; 191.3°
5.0 m	16.5°; 16.5°	32.0°; 32.0°	63.8°; 63.8°	127.8°; 127.8°	191.8°; 191.8°	255.8°; 255.8°	455.6°; 95.6°
2.5 m	8.3°; 8.3°	16.0°; 16.0°	31.9°; 31.9°	63.9°; 63.9°	95.9°; 95.9°	127.9°; 127.9°	227.8°; 227.8°
1.5 m	5.0°; 5.0°	9.6°; 9.6°	19.1°; 19.1°	38.3°; 38.3°	57.5°; 57.5°	76.7°; 76.7°	136.7°; 136.7°
0.5 m	1.7°; 1.7°	3.2°; 3.2°	6.4°; 6.4°	12.8°; 12.8°	19.2°; 19.2°	25.6°; 25.6°	45.6°; 45.6°

Measured for 50cm coax cable: 1 MHz -1deg/ 15MHz -15deg./30MHz-30deg

1.8MHz 2.6ns, 15MHz 2.7ns, 30MHz 2.3 ns

3.0 The NCC-2 (DXE) Receive Antenna Phasing Controller

Front Panel Controls and Switches



In the NCC-1 there is no RTR-Interface, no pre-amp and *both* CH A and CH B can be phase inverted. Consequently, on the rear panel there is no PTT but T/R CNTR to protect active RX antennas from damage by TX energy (cutting off DC supply voltage).

Rear Panel NCC-2



The NCC-2 is therefore connected externally between the Power Amplifier (or just main TX Ant.) and the Transceiver TRX.

3.1 Controller Functionalities

Fast RTR-RX/TX (PTT) switching, Norm= NCC-2 (Phasing) active, CH A, CH B, attenuation (0/-10/-20/-30 dB), balance (-10-0-10dB), pre-amps +16dB, phase inverting, Band L (160/80m) /H 40m to 10). The *design impedance* for input/out ports is 50 Ohm.

There are principally *two operational modes* possible:

Each Channel (A, B) can prudently be equipped with signal RX safe-level limiter and the pre-amps as an option. If needed a transformer (75/50Ohm) can go before CHA/B in.

1. RX Antenna Array Phasing
2. RX Noise Reduction (mainly from near-by QTH environment, because in DX ionosphere sets the phase momentarily in an uncontrollable manner). A 50cm coax jumper connects Main Ant out to CH A (Main Ant.) in.

The TRX antenna carries good and bad signals, the auxiliary antenna desirably only the local EMI. The RX Ant. Phasing Controller vectorially subtracts the bad noise signal and improves within limits the RX SNR. In theory this works only well for one single EMI point source.

3.2 Practical limitations

Cancellation is narrow in “spatial signature”

A single phase-and-amplitude setting supplies one complex cancellation coefficient. It can deeply cancel one dominant *coherent* source, or multiple signals only if they happen to have nearly the same ratio between the two antennas.

If two noise sources have different spatial relationships:

$$\frac{N_{1A}}{N_{1B}} \neq \frac{N_{2A}}{N_{2B}}$$

one setting generally cannot cancel both deeply.

That is why analogue phasing cancellers are less effective against numerous distributed broadband sources.

The null is frequency dependent

Cable delays and physical path delays produce phase changes proportional to frequency:

$$\phi = 2\pi f\tau$$

If the phase is perfect at one frequency, it may not remain perfect over a very wide bandwidth. A broadband digital signal or switching-noise comb can therefore show excellent cancellation on one part of the spectrum and less cancellation elsewhere.

The reference must be coherent

The noise on the sense antenna must be *sufficiently correlated* with the same noise on the main antenna. Antenna polarization matters. Completely independent noise, atmospheric noise or noise generated after the combiner cannot be cancelled this way.

Amplitude accuracy matters as much as phase

If two noise voltages are exactly opposite in phase but differ in amplitude, a residual remains. For example, a *1 dB amplitude mismatch* limits the theoretical cancellation to *roughly 25 dB* even if the phase is exact. Likewise, a phase error of only a few degrees can materially reduce a deep null. This is why the NCC-2's large, repeatable controls and wide phase range are more than ergonomic luxuries and may even need mechanical vernier control. They allow finer vector matching than many simple QRM-eliminator circuits.

4.0 Testing NCC-2

4.1 NCC-2 Lab Tests and difference to NCC-1

NCC-2 and NCC-1 details are listed on

<https://www.dxengineering.com/parts/dxe-ncc-2>

[https://www.dxengineering.com/parts/dxe-ncc-](https://www.dxengineering.com/parts/dxe-ncc-1?srsId=AfmBOoqP93QlxFycws2hGb6caMf3gMx0Qf5RetiG2jgvCJSpMNaj4dRc)

[1?srsId=AfmBOoqP93QlxFycws2hGb6caMf3gMx0Qf5RetiG2jgvCJSpMNaj4dRc](https://www.dxengineering.com/parts/dxe-ncc-1?srsId=AfmBOoqP93QlxFycws2hGb6caMf3gMx0Qf5RetiG2jgvCJSpMNaj4dRc)

There are additionally several "in action" YouTube videos on the net.

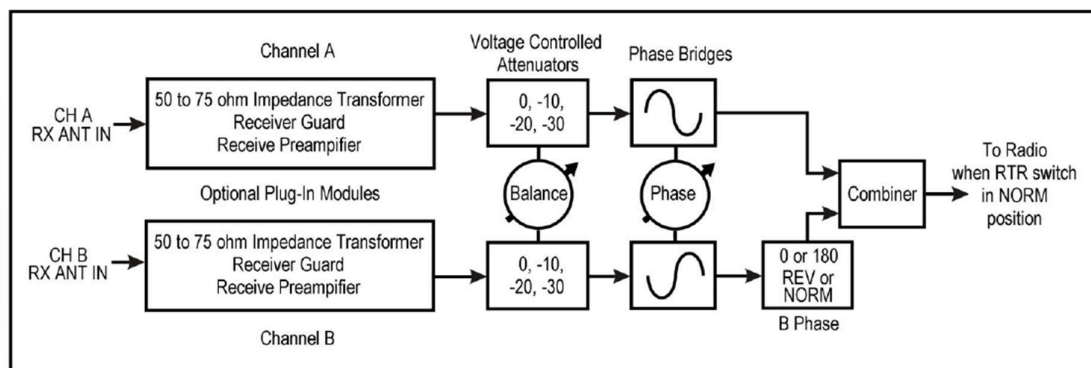
There was a review by VA2PV, May 2024 QST, "DX Engineering **NCC-2** Receive Antenna Phasing Systems" (Preamp (16dB) /Limiter built in (max 10W in), nearby low, small RX Ant.) with physical description, connections and On-the-Air Operations using YAESU FTDX101MP. Result: S9+20dB Noise nulled to almost nothing on 40m.

An NCC-1 System review by N0AX appeared in Aug. 2014 QST, "DX Engineering AAPS3-1P Receiving Array"

In our Lab test we will focus more on the technical/measurement capabilities/limits of the units.

From a practical use of NCC-2 **as a noise reduction unit** the low bands 160/80m/40m will mostly be of primary interest because the man-made noise (EMI/RFI) typically decreases from 160 to 40 to 10m. Sometimes however, very local interference sources matter on the higher bands.

The block diagram of NCC-2 and NCC-1 are basically identical.



4.2 Test Report -Device under Test definition- Test Equipment

We investigated NCC-2/NCC-1 in the following configurations:

- with main board 2016 WP0050 Rev.1
- back panel RTS rev. 1.0 WP0052
- back panel RTP-2 rev. 2.0 WP0052
- front panel NCC-2 FP rev. 1.2 WP0048
- DXE RX Guard RG5000HD, in Ch A, in CH B
- DXE Pre-Amps in CH A, in CH B

The phase bridge design/ transistors changed considerably over NCC-1.

The function of the former DXE RTR-2 modular RX/TX Interface is now integrated.

We investigated NCC-1 (2014) in the following configuration:

- with main board 11/25/08 Rev.2b DXE
- front panel NCC- FP rev. 2 10/17/06 DXE

Testing was carried out between 02 July 2026 and 23 Aug. 2026 at HB9CVQ

Room temperature was 24 to 29 C.

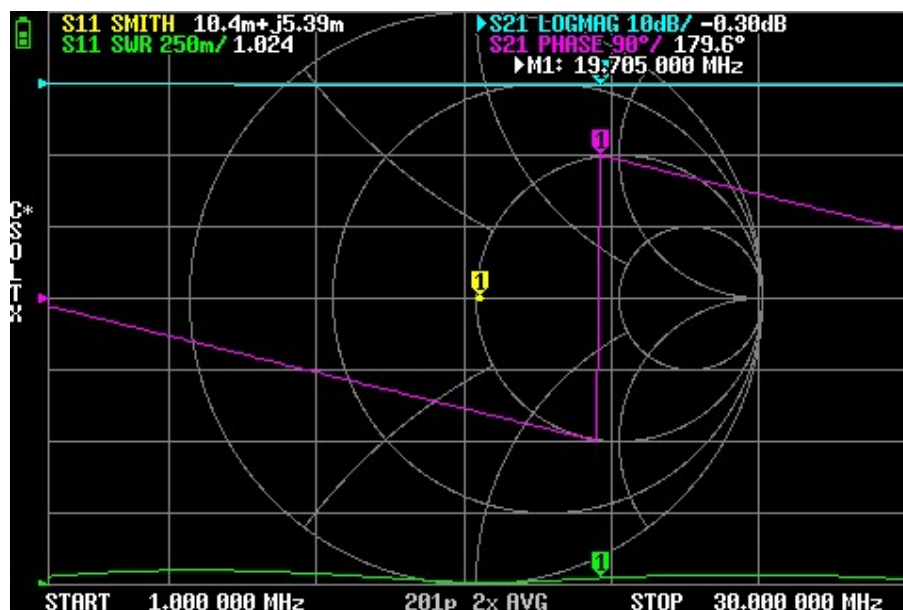
The following cross-check calibrated test/*measurement instruments (mostly low budget)* were used:

1. Rigol Function Generator DG1032Z (up to 30MHz)
(CH1/CH2 50Ohm, phase coherent, adjustable delay, sinus/noise)
2. Huber/Suhner power splitter 50Ohm / -6dB (GHz)
3. Various BNC 50Ohm attenuators (fixed/variable)
4. 50Ohm Hybrid power splitter (2 x -4dB, 30MHz) better port isolation than (2.)
5. LiteVNA 64 (50k to 6.3G)
6. TinySA-ultra (100k up to 5.4G)
7. Siglent SVA101X Spectrum/VNA (9K-1.5G)
8. True RMS Multimeter UNI-T UT171B
9. HP 8647A Signal Generator
10. Tektronix Oscilloscope 2465A (2-CH 350MHz)
11. Elecraft XG-3 RF-Signal Source 50Ohm
12. Rig Expert AA 600 (TDR check)
13. FTDX101MP, SW/FW 2025

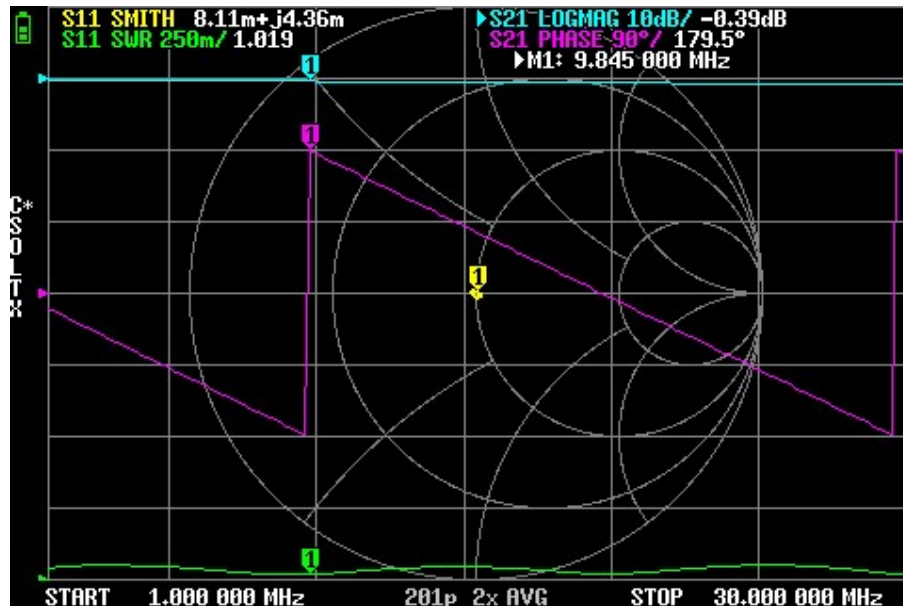
4.3 How do system relevant coax-cables behave in terms of phase shifting

The VNA graphs show VSWR (green), Smith Chart S11 (yellow, matching), S21 Through Losses (with amplitude "blue" and phase "magenta") over a frequency range of 1 to 30MHz. The marker M1 is set to the 180 deg phase shift frequency. The jump in phase from minus to plus is a display (measurement) issue, not a physical effect. The phase does not jump but continues smoothly from 0 to 180 deg portion to 180 to 360 deg and so on (**phase wrapping** or **phase rollover** / **phase ambiguity**). RG223 is just double shielded RG58C/U coax. Coax is terminated with 50Ohm from VNA input.

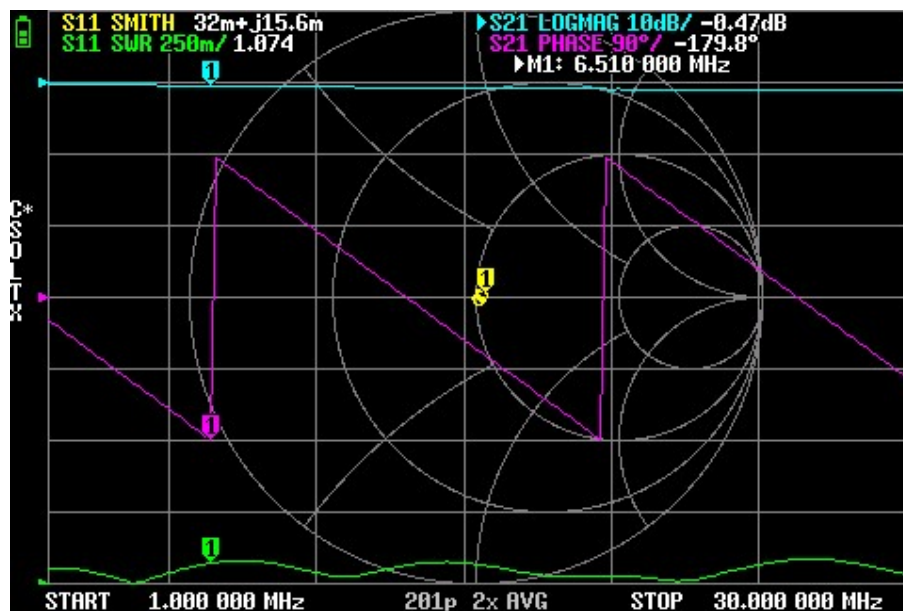
RG223 (50Ohm, 0.66) 5m long (M1=19.7MHz)



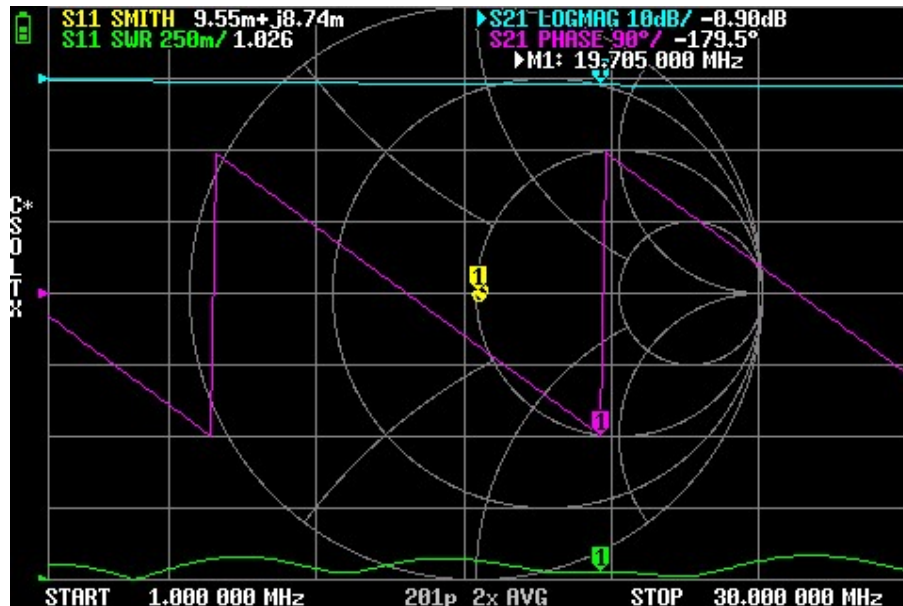
RG 223 (50Ohm, 0.66) **10m** long (**M1=09.8MHz**)



RG 223 (50Ohm, 0.66) **15m** long (**M1=6.5MHz**, first block of 180 deg phase shift)

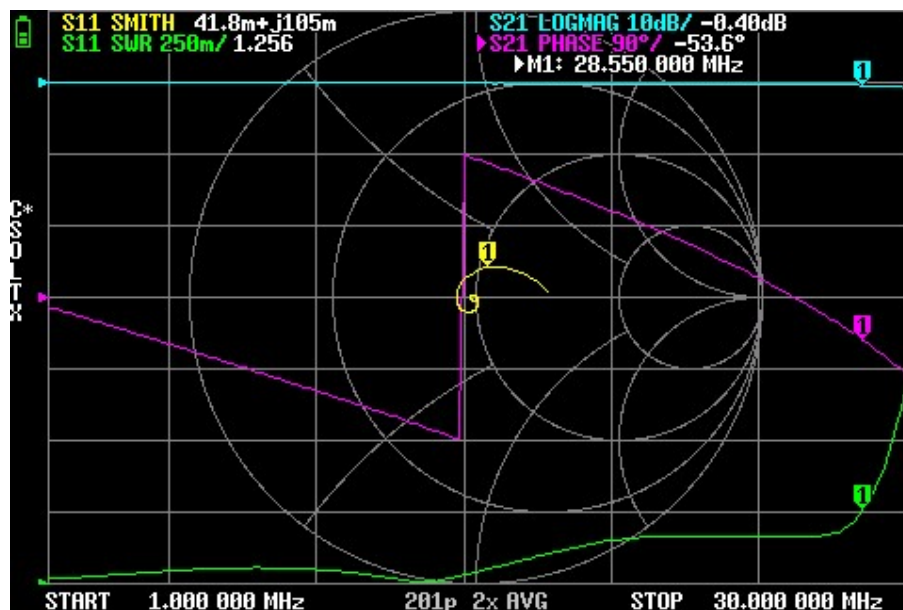


RG 223 (50Ohm, 0.66) 15m long (M1=19.7MHz, second block of 180 deg phase shift)



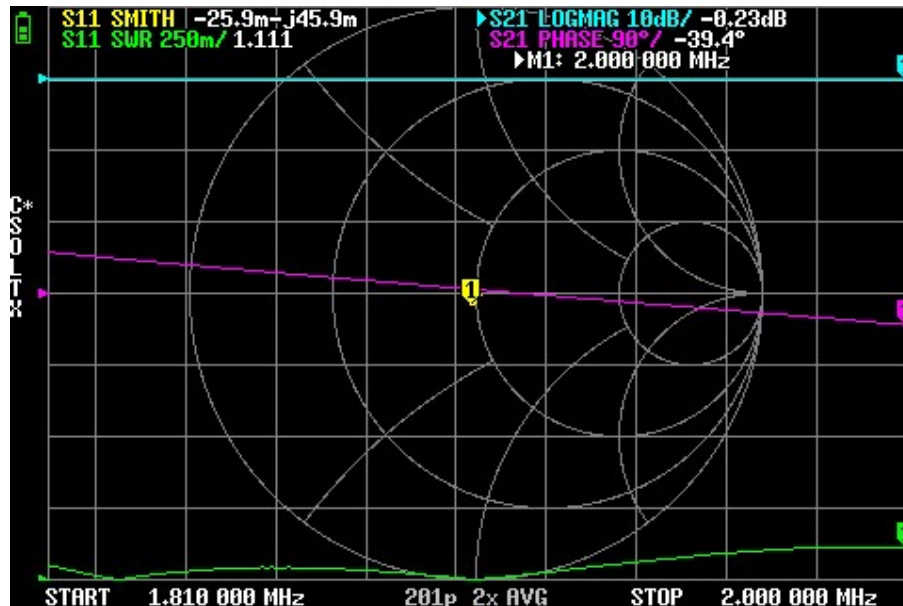
4.4 How do system relevant filters behave in terms of phase shifting

Low Pass Filter Drake TV3300LPF --41MHz/-80dB-- (Span 1 to 30MHz, Marker 28.5MHz)



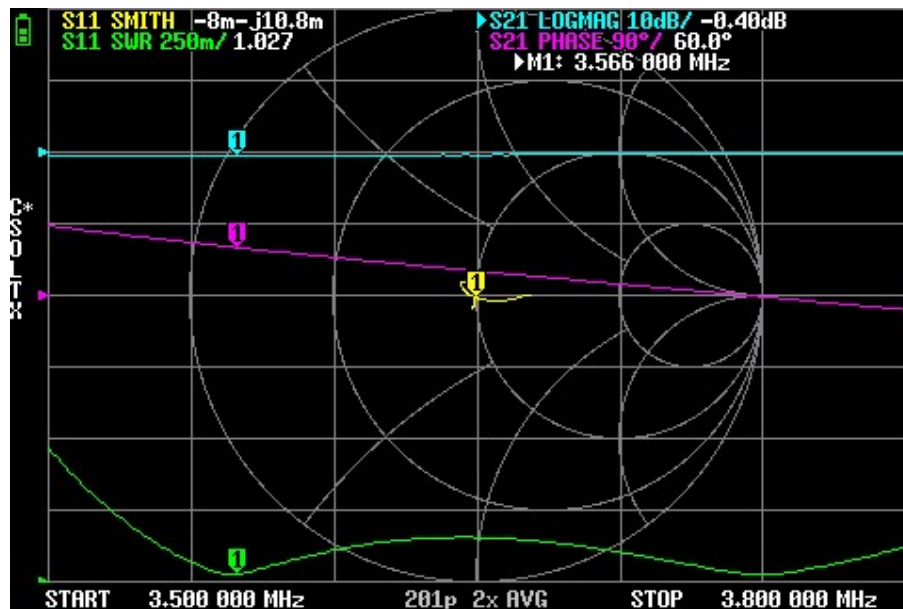
Well match, low losses, continuously shifting phase as frequency goes up

160m HAMATION Band Pass Filter (Span 1.8 to 2MHz, Marker 2MHz)



Since this is a 160m Band Pass Filter there is little phase shift (ca. +/-40 deg.)

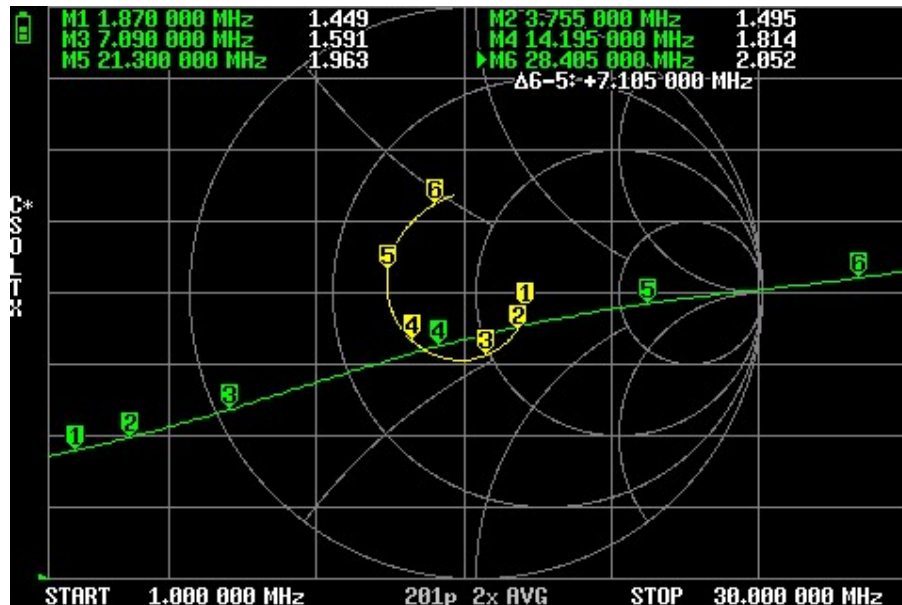
80m HAMATION Band Pass Filter (Span 3.5 to 3.8MHz)



Since this is a narrower 80m Band Pass Filter there is relatively small phase shift

4.5 Back to NCC-2 RX-Ant. Phasing Controller (Input)

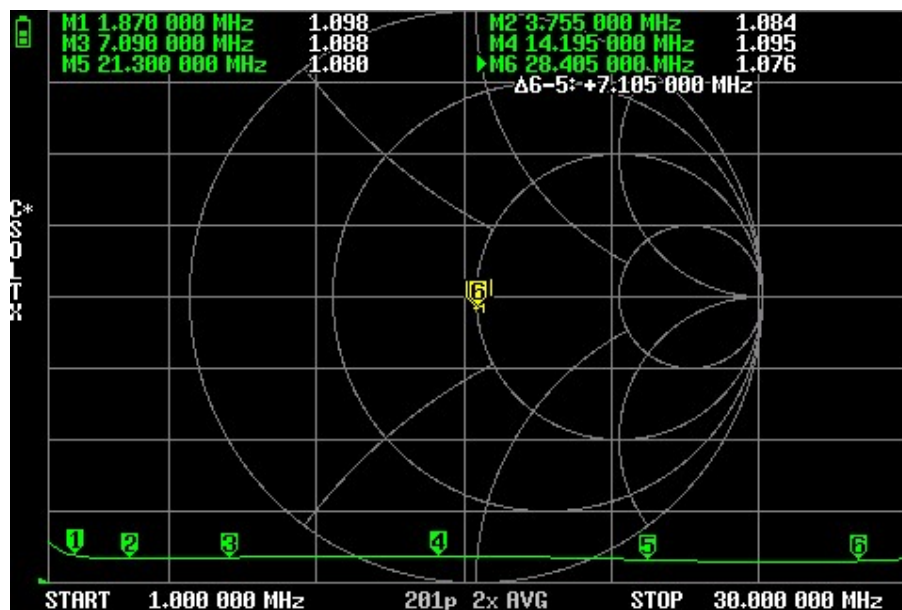
NCC-2 Input VSWR CH A, Att.=0dB selected (Span 1 to 30MHz)



VSWR remains below 2 (ca. 11% power reflection) **M1= 1.8MHz, M6 =28.4MHz**

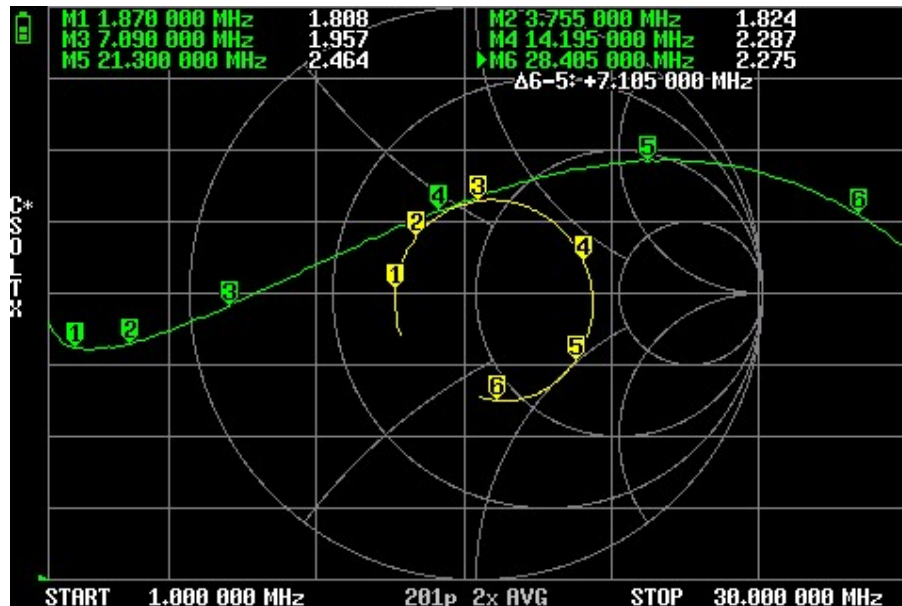
There is no forced/switched on input active attenuation (Att.=0dB)

NCC-2 Input VSWR CH A, Att.=10dB selected (Span 1 to 30MHz)



Very good matching (ca. 20dB RL)

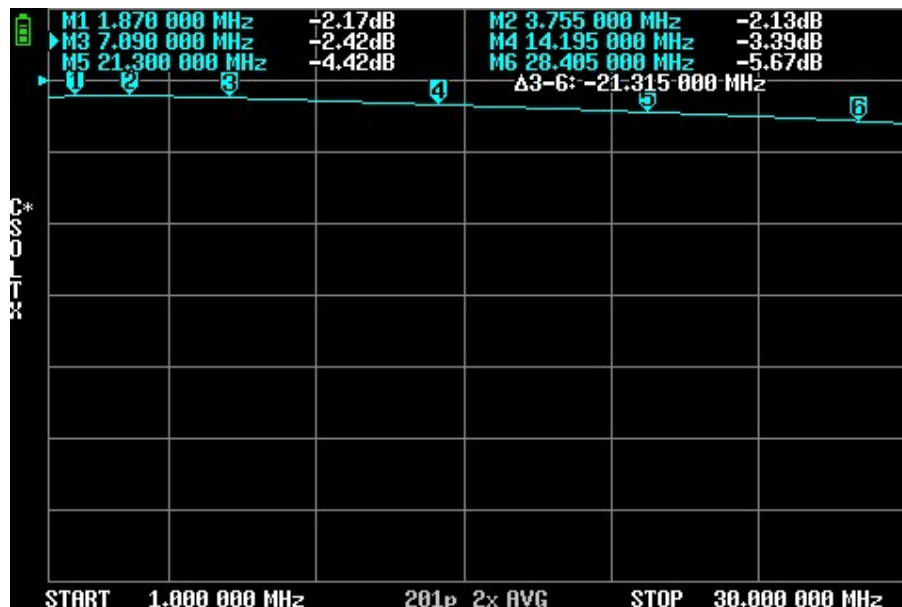
NCC-2 Input VSWR CH A with +16dB pre-amp on (Span 1 to 30MHz)



A small VSWR degradation compared just bare CH A with Att.=0dB, still around SWR 2

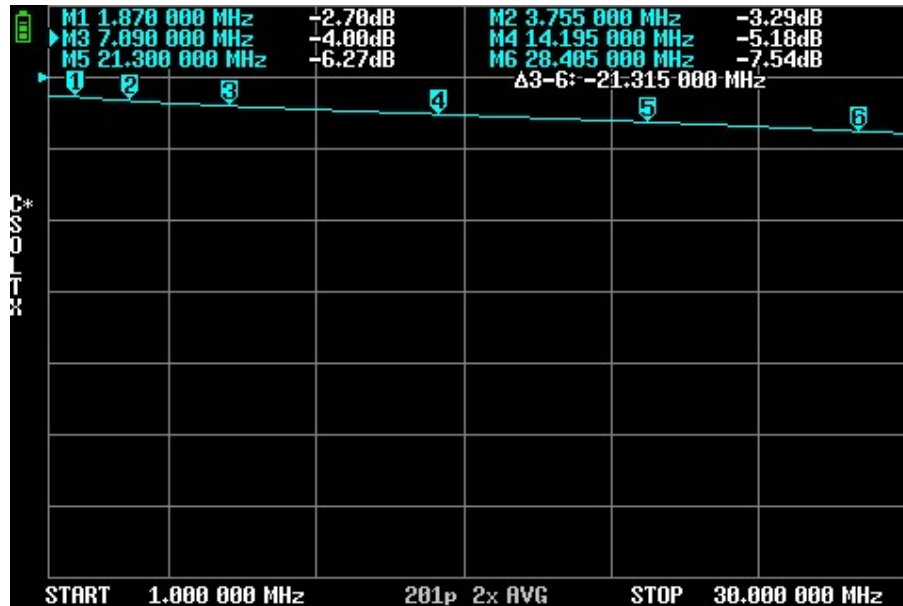
4.6 Back to NCC-2 RX-Ant. Phasing Controller (Through CH-Losses)

NCC-2 CH A -Through Losses in Pos. L -for 160/80m- (Span 1 to 30MHz)



ca. 2 to 6 dB Input-> Through losses at Output (Switch Pos L -low bands)

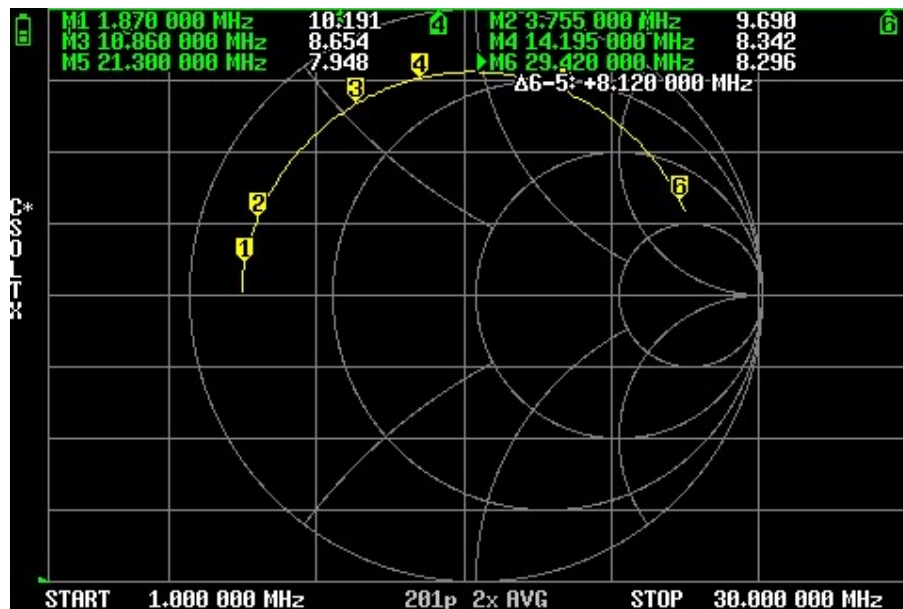
NCC-2 CH A - Through Losses in Pos. H - for 40/10m- (Span 1 to 30MHz)



ca. 3 to 8 dB Input-> Output losses (Switch Pos L -low bands)

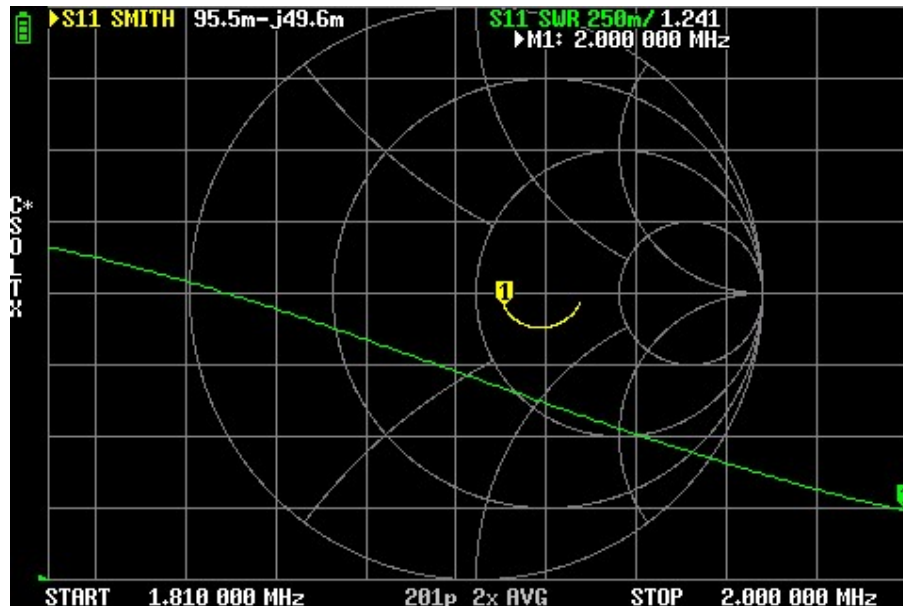
4.7 Back to NCC-2 RX-Ant. Phasing Controller (Output Matching)

NCC-2 Output (to TRX) VSWR (Span 1 to 30MHz)



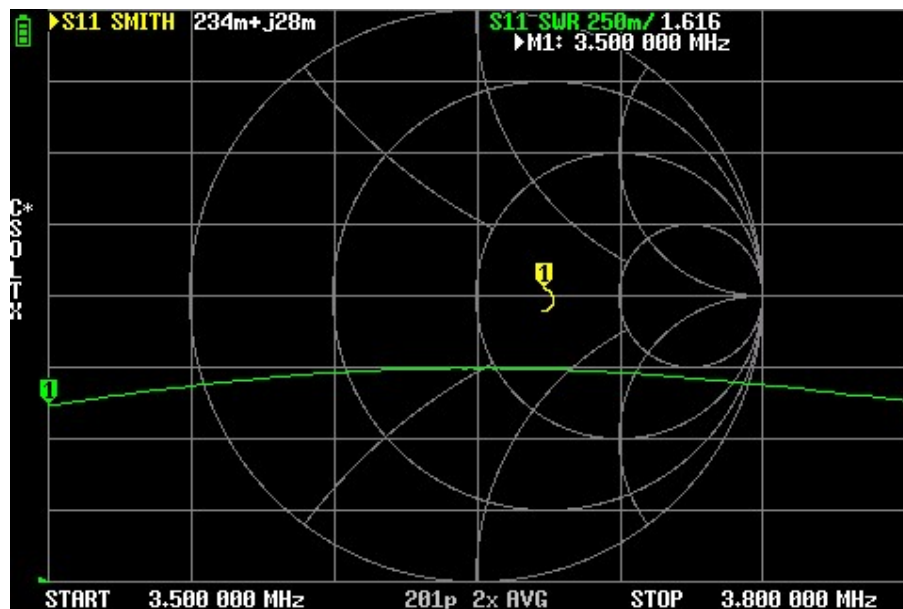
Considerable output mismatch (no 50Ohm) to TRX (1.8MHz SWR =10!),

YAESU FTDX101MP -Year 2019- (IPO), no preamp, 160m VSWR at RX input



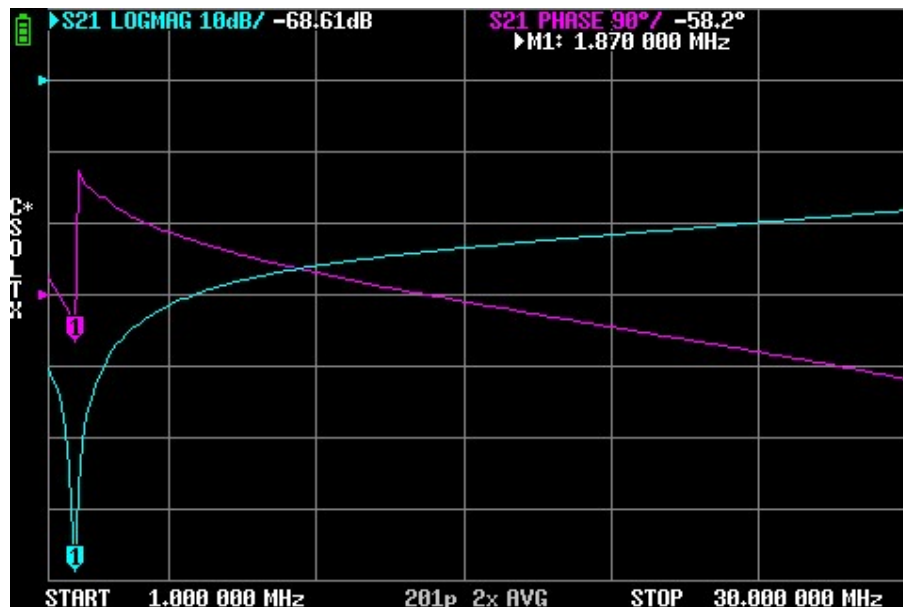
Luckily good match (SWR 1.2) at the used TRX input side. RX Input settings with Amp1, Amp2 showed not so good matching of up to SWR 5. Similar effects were apparent in Elecraft K3S, in contrast to excellent match in SDR Perseus. A mismatch of VSWR 2 may lead to worst case error of 6dB, VSWR 3 can cause 10dB IARU calibrated S-Meter errors.

FTDX101MP (IPO) 80m VSWR at RX input, a little worse



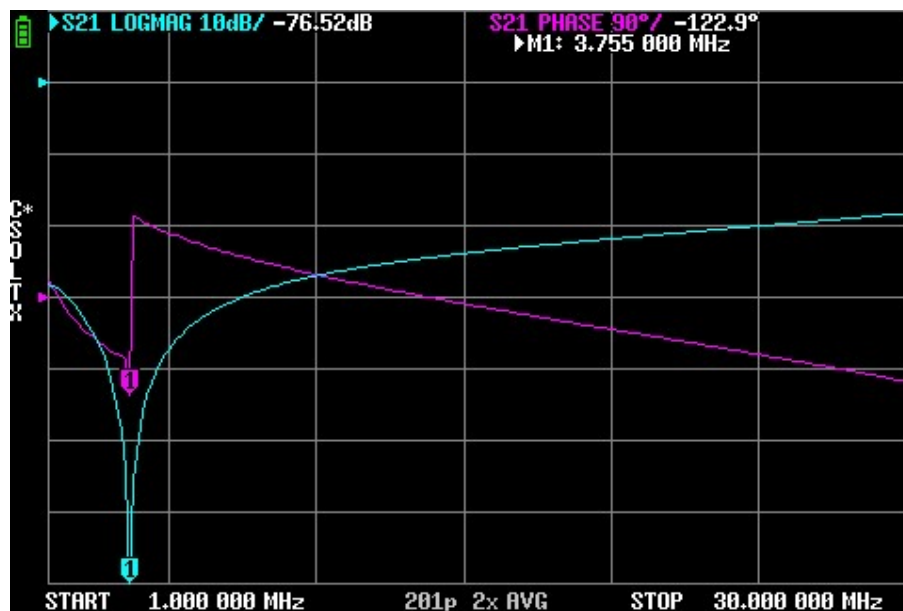
4.8 Back to NCC-2 RX-Ant. Phasing Controller (Notching)

NCC-2 160m Notch, Pos L



Here the Notch depth is around 70dB in the Lab, very sensitive to adjust

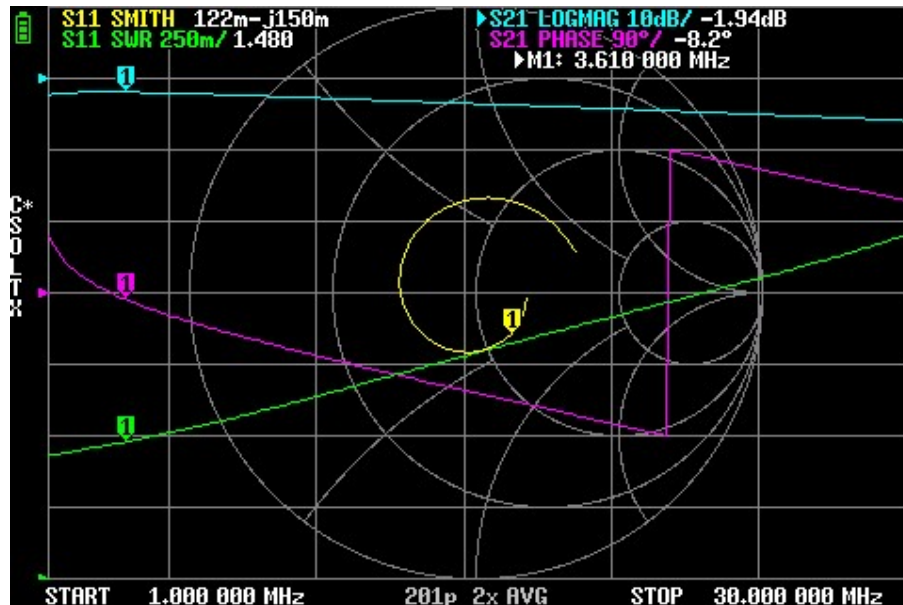
NCC-2 80m Notch, Pos L



Here the Notch depth is around 77dB in the Lab, very sensitive to adjust

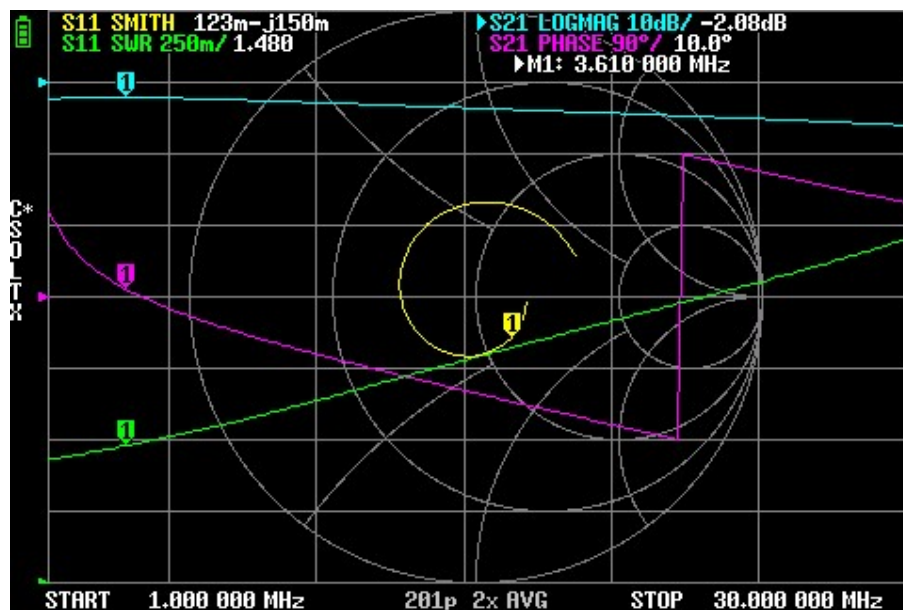
4.9 Back to NCC-2 RX-Ant. Phasing Controller (Notching Limits)

NCC-2 Notch: **CH A**, Pos L, Att.=0, **Phase max left**, Marker 80m (Span 1-30MHz)



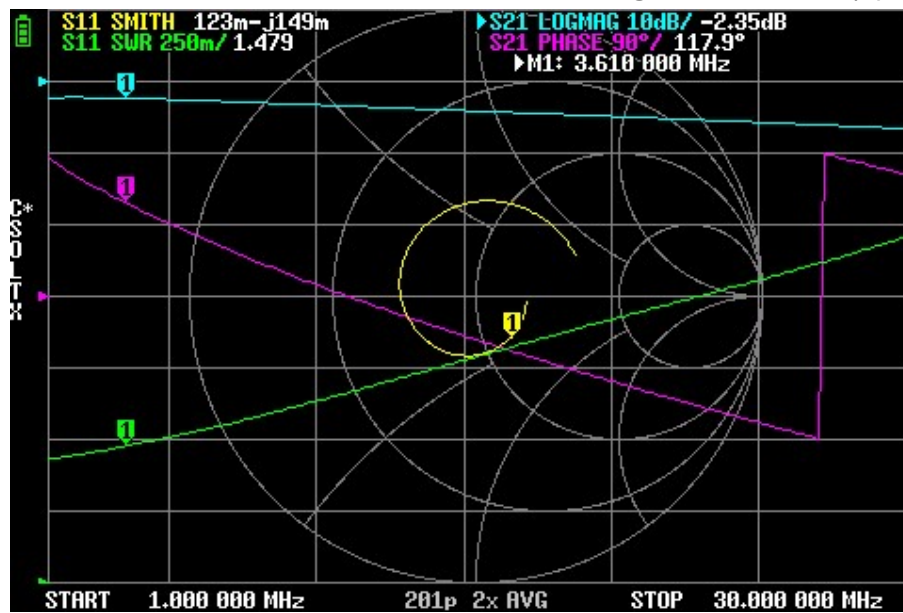
Phase -8 deg.

NCC-2 Notch: **CH A**, Pos L, Att.=0, **Phase mid**, Marker 80m (Span 1-30MHz)



Phase +10 deg.

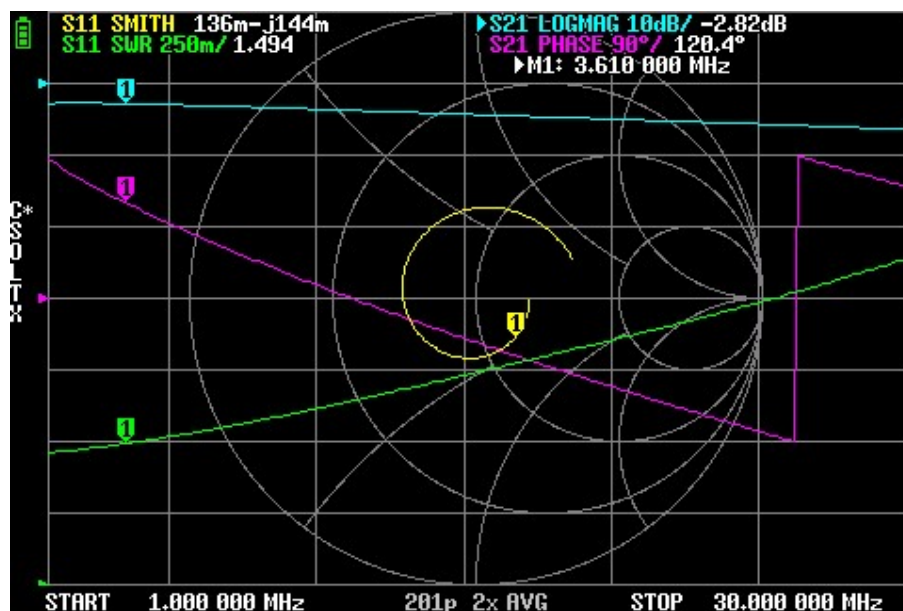
NCC-2 Notch: **CH A**, Pos L, Att.=0, **Phase max right**, Marker 80m (Span 1-30MHz)



Phase +118 deg.

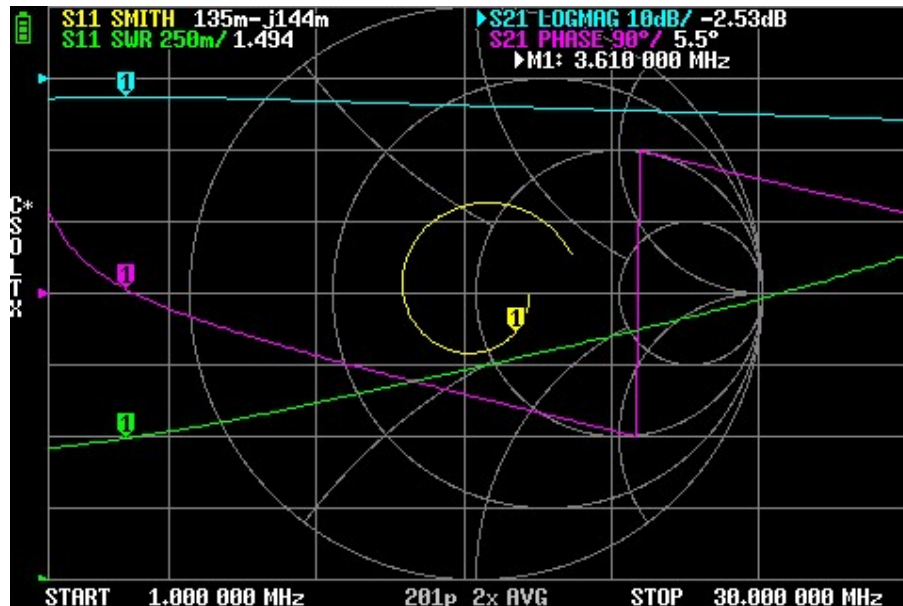
Now follows NCC-2 phasing for CH B

NCC-2 Notch: **CH B**, Pos L, Att.=0, **Phase max left**, Marker 80m (Span 1-30MHz)



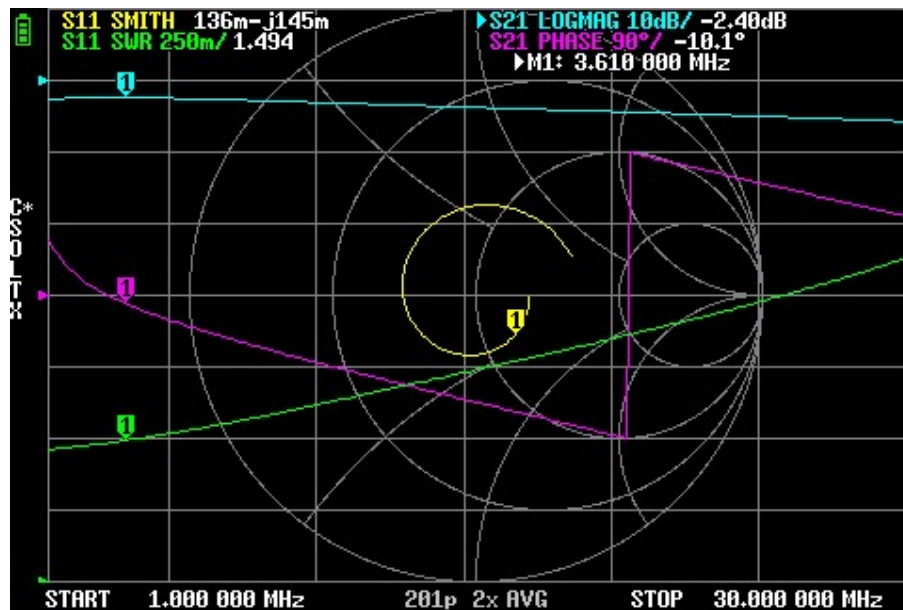
Phase +120 deg.

NCC-2 Notch: **CH B**, Pos L, Att.=0, **Phase mid**, Marker 80m (Span 1-30MHz)



Phase +6 deg.

NCC-2 Notch: **CH B**, Pos L, Att.=0, **Phase max right**, Marker 80m (Span 1-30MHz)



Phase -10deg.

- BTW: The noise figure of the NCC-2 preamp (16dB) is around 4dB on 17m.
- BTW: The preamp has ca. 10ns Group Delay, like a 2m RG58C/U. Therefore +16dB and -10dB ATT. -6dB BAL compensate to 0dB. This can be used above 40m to correct the phase shift (end) range if needed. The attenuation can be fine-tuned with BALANCE-with very little phase shift impact.

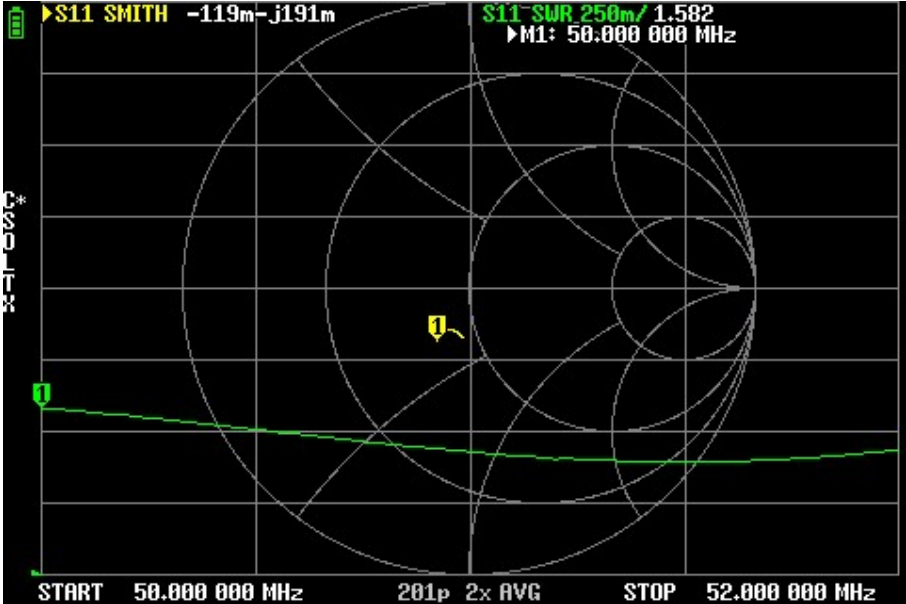
4.10 Tested Notch depth NCC-2 VNA method is close to 2 port Rigol Generator

In a rough overview:

160m	-70dB
80	-76
40	-70
20	-65
10	-70

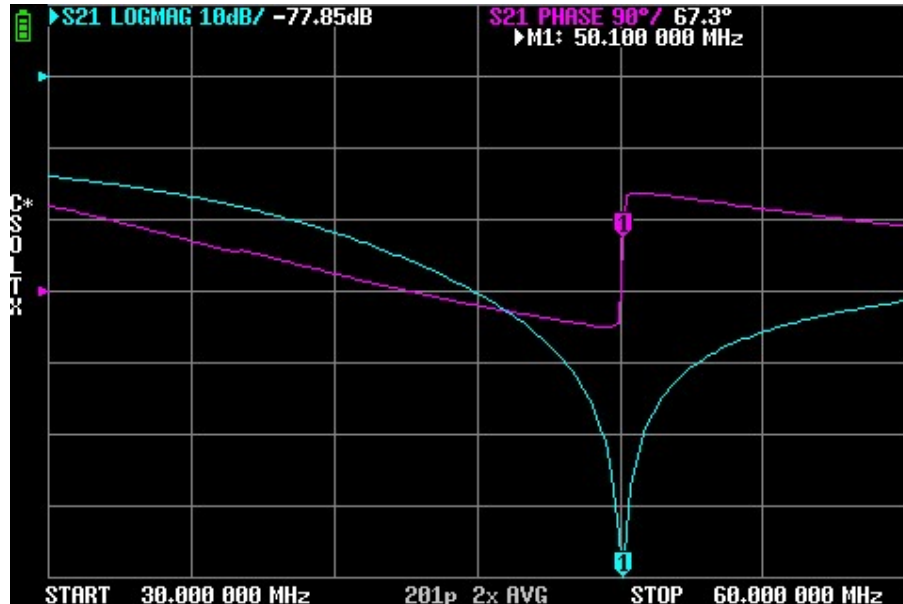
4.11 Unexpected NCC-2 performance on 6m

NCC-2 Input VSWR CH A (M1 50.0 MHz)



Still good VSWR Input match on 6m

NCC-2 Notch on 6m band (M1 50.1MHz)



Surprisingly good Notch depth with 78dB on 6m, but phase adjustment range is reduced, not shown here.

4.12 NCC-2/NCC-1 dual port signal generator testing

Test equipment used:

Rigol Function Generator DG1032Z (up to 30MHz, CH1/CH2 50Ohm, phase coherent, adjustable delay, sinus/noise)

Huber/Suhner power splitter 50Ohm / -6dB (GHz)

Various BNC 50Ohm attenuators (fixed/variable)

50Ohm Hybrid power splitter (2 x -4dB, 30MHz) better port isolation than R-splitter

4.12.1 Signal Overload Testing of NCC-2 Frontend

Due to given instrumentation limitations, we could still however prove with a restive as well as hybrid power splitter (better port isolation) in a two tone IMD3 test on 80m (up to $S9+50\text{dB} \Rightarrow -23\text{dBm}@50\Omega$) with the FTDX101MP (IPO, ATT.=0dB, CW 300Hz and ALC=off), compare <http://www.sherweng.com/table.html> , that the NCC-2 does not degrade FTDX101MP RX overload performance.

4.13 NCC-1 phasing range (DC=13.8V)

NCC-1, Att. A and B =0dB, Balance =Mid, (no 50cm jumper-coax was used, no preamp)

1.87 MHz	Phase max. left	Phase mid	Phase max. right	Phasing
	+127 deg	+117 deg.	+78 deg.	Norm B, H
	+85	+68	+37	Norm B, L
	-50	-64	-102	Rev. B, H
	-93	-113	-144	Rev. B, L
28.405 MHz	-158	-168	-166	Norm B, H
	-172	-173	-174	Norm B, L
	+14	+11	+6	Rev. B, H
	0	-1	-3	Rev. B, L
2xRev. A and B	-158	-9	-122	Rev. A, H
	-169	-168	-155	Rev. A, L

BTW: CH A to CH B present from 1 to 30 MHz ca. 60dB port isolation.

4.14 NCC-1 Notch Depth

NCC-1 shows between 160m to 20m ca. 40 to 50dB (extremely sharp), below on 15/10m very tricky to find any dip. On 6m maybe best case 8 to 11dB.

Input VSWR is mostly under 2.5, Output VSWR not measured.

On 10m (CH A and CHB) 0/10/20/30dB Att. claim reveals 4.3/14.7/25/35 dB measured attenuation.

5.0 NCC-2 Noise canceller used in System at QTH installation HB9CVQ

After the lab tests it is now time to face reality in my QTH with the NCC-2 installed in a system.

My two antennas are 2x36m@24m doublet, remotely tuned with ATU under roof, from 160 to 10m, as sloping inverted Vee (ca. East-South, 100deg H, 45 deg V) configuration. Additionally, there is the SteppIR Yagi DB18E @19m above ground, on a rotating mast,

above the tile roof of the house. On 40m acting as shorted 2 el Trombone Yagi, element spanning ca. 11m and 6m long boom. The shack is on the second floor, right below the attic.

A horizontally attic mounted ALA 1530 Wellbrook 1m-diameter, active magnetic loop antenna serves as noise pick-up from 50kHz to 30MHz. This antenna is remotely biased over RG58C/U with many distributed common mode chokes along the feeder.

There is a DXE signal limiter in the shack and in each NCC-2 CH A and B.

The decoupling (40/20/10m) of the TRX antennas (Yagi-inv. Vee) is listed in the following table. Yagi pointing by rotation to from SW to NW.

Yagi to inv. Vee Antenna Decoupling

	40m	20m	10m
210 deg South America LU	-28 dB	-29 dB	-43 dB
240 PY	-21	-25	-54
275 Central America	-21	-24	-41
305 USA east coast	-23	-25	-35
328	-28	-21	-35



HB9CVQ QTH

inv. Vee in red above DB18E Attics with portable EMI TX loop noise source



Portable Noise Source set-up in basement position to simulated interference, a local QRM source, well EMI filter decoupled (60dB 1-30MHz). Very close (10cm) to the split in top-shield of the loop we measured about 1V/m.

inv. Vee/Yagi to 1m horizontal ALA 1530 decoupling (noise pick-up, Low Pass Filter Drake TV3300LPF cut off 41MHz/-80dB in shack) Test done right between coax-cables connected to Main and Auxiliary Ant. (mag. loop) at NCC-2. Yagi is positioned to 305deg USA.

160m inv. Vee	1.81 MHz	-38dB
80m inv. Vee	3.51	-44
40m Yagi	7.001	-42
40m Inv. Vee		-51
20m Yagi	14.001	-35
20m inv. Vee		-45
15m Yagi	21.001	-27
15m inv. Vee		-41
10m Yagi	28.001	-30
10m inv. Vee		-43

Now we installed the portable Noise Source

Rigol Wave Generator, 40dB, 0.8 to 1000MHz broadband 50Ohm 3W ENI amplifier,
Drake LP Filter 3dB cut off 41 MHz, -6dB 50 Ohm power attenuator (forced matching),
TEKBOX small magnetic TX loop antenna

<https://emcsupplies.com/collections/antennas/products/tbma6-p-passive-loop-antenna>

Table giving the CW noise source details (Attic, 2. Floor, Basement, signal levels, Notch)

Frequency	RX level Main Ant	RX level Aux. Ant	Best Notch in dB
1.819 MHz EMI Source in Attic	S9+6dB	S9+6dB	30dB
3.519 EMI source in Attic	S9+3	S7	40
7.019 EMI source in Attic	S9	S9+10	>30
14.019 EMI source in Attic	- test failed	-	-
28.019 EMI source in Attic	S2 (with AMP2 on)	S2 (with AMP2on)	35

FTDX101MP AMP2 on = +20dB over IPO, 10dB over AMP1

Table demonstrating attenuation between EMI source positions/floors

EMI Location	Attic	Floor 2	Basement
1.81 MHz	-55/-66dBm*	-64dBm	-75/-76dBm*
3.5 MHz	-58/65	-60	-78
7.0 MHz	-61	-67	-81
14 MHz	-55	-75	-83
28 MHz	-60	-73	-81

*Polarization/Rotation of EMI Loop dependent

Listed EMI levels were all measured with SDR Perseus in CW BW, Preselector on, Dithering on.

EMI is not only purely radiated by field but also effected/spread by conducted transport (powerlines LVD/ telecom lines).

Secondary radiators and potential mains cable resonances play a role. There is no such thing as a clear building attenuation. Wires/installation play a critical, individual role.

At least for 160/80 and 40m the house wiring is in the near field of the portable EMI local QRM source. Lambda/6 is a very rough estimate.

Summary:

Motivated by the ever-increasing HF noise floor on the HAM Bands and considering all reasonable local shack denoising measures, there is maybe still some EMI to be Noise-Cancelled by analogue 2 channel -180 phase reversal (DXE NCC-2). The options to the procedure and the history/patents of HF phase reversal have been analyzed.

The basic and necessary success requirements (e.g. main antenna must see a coherent noise-copy of auxiliary/noise-pick-up/ antenna) are discussed in detail. System efficiency Check-Out procedures and recommendations are presented.

NCC-2 and the older NCC-1(before 2016) are compared and are tested in the lab. This includes Input/Output Port VSWR matching from 1 to 30/50MHz, Throughput Attenuation and Phase Range adjustment limits.

The **NCC-2 in the Lab Test reaches over 70dB Notch** (very sharp tuning) capability from **160 to 6m. NCC-1** shows between **160m to 20m max. 40 to 50dB** (extremely sharp). Below on 15/10m it is very tricky to find any dip. On 6m best case is 8 to 11dB.

System relevant tests for antenna coupling were performed at HB9CVQ QTH. There is a remotely tuned inv. Vee doublet (160 to 10m@24m over ground, ca. 11m above roof) and a SteppIR Yagi DB18E-40-10m (@19m over ground, 6m above tile roof)

The decoupling between the two TX ants. to RX noise pick-up, active 1m Magnetic Loop ALA-1530 was measured between 160 to 10m.

A special, portable, local-QRM noise-set-up was designed to simulate a single source of EMI interferer nearby. This electrically small, forced 50Ohm matching, 60cm diameter, shielded, passive CISPR loop with Lowpass-Filter was powered from a well EMI mains decoupled broadband 3W amplifier and fed from a signal generator for 160/80/40/20/10m performance.

Three **local QRM house-locations were system tested: Attic, 2. Floor, Basement.** The carefully tuned **NCC-2 band notches resulted in ca. 30 to 40 dB** (6dB=1S-unit acc. IARU)

The effect of several, simultaneously occurring Noise Sources could not be tested systematically and in detail, yet.

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